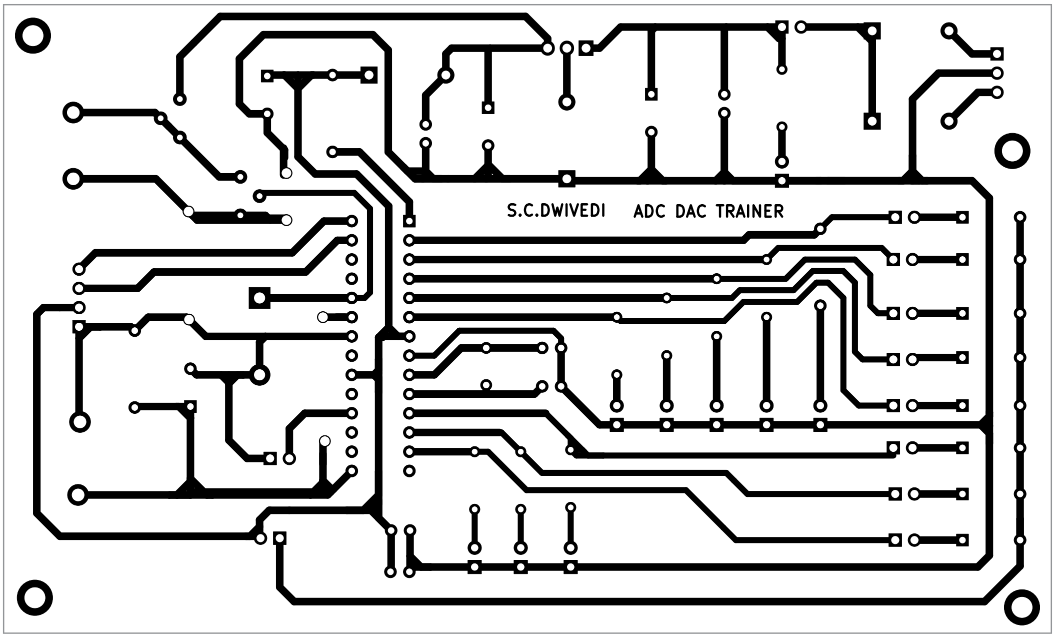

Fig. 2: Actual-size PCB of trainer

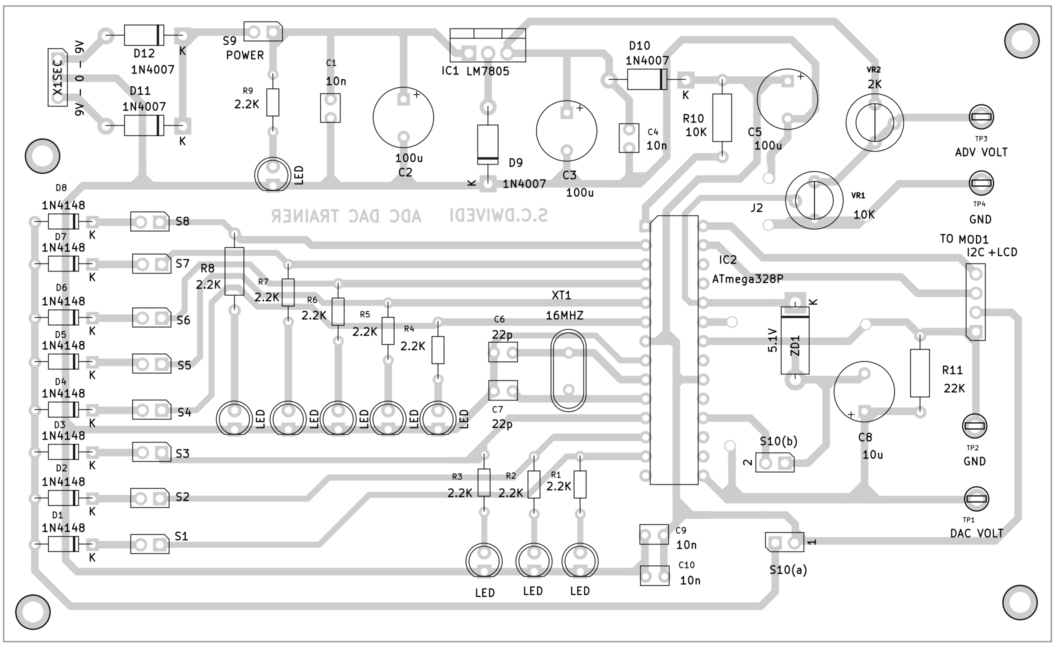

Fig. 3: Component layout of the PCB

ADC input is connected from 5.6V supply rail to calibrate the maximum ADC input to exact 5V using pot VR2. Zener diode ZD1 is connected as protection for the ADC pin.

The circuit mainly works in two modes, ADC and DAC. When switch S10 is on, the device operates in DAC mode. When the switch is off, it toggles to ADC mode.

Diodes D1-D8 are used to show exact data generated by the microcontroller in ADC mode. Improper selection of switch S1-S8 in this mode may cause to light up improper LEDs. The protection diodes D1-D8 prevent the supply from microcontroller to the common supply rail of DAC input and ensure proper LEDs light up.

The I2C module and 20x4 LCD are used for the hardware simplicity. The controller SCL and SDA pins (A4 and A5 of I2C) are connected to PC5 and PC4 pins of IC2. The ATmega 328P runs at oscillator frequency of 16MHz because of crystal XT1 and capacitors C6 and C7 act as decoupling capacitors.

Resistors R1-R9 are current-limiting resistors for respective LEDs. LED9 is for power-on indication and switch S9 is the power on/off switch. Resistor R11 and capacitor C8 convert the PWM DAC output to normal DC voltage. As per the data sheet of ATmega 328P, capacitors C9 and C10 should be connected as near as possible to the power pins of the controller chip during assembly.

The software is written in Arduino C language Sketch and is well explained in the code. The software uses a special technique of employing a single port for input and output, which may be called port manipulation. Special libraries used in the software are Liquid cystalI2C.h and wire.h. Their installation process is available at online sites.